

Experiment 9

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Aim

To create and implement PL/SQL packages by developing a package specification and package body containing procedures and shared cursors, in order to achieve modular, reusable, and efficient database programming.

Software Requirements

- Database Management System:
 - Oracle Database Express Edition (Oracle XE)
 - PostgreSQL Database
- Database Administration Tool / Client Tool:
 - Oracle SQL Developer (for Oracle XE)
 - pgAdmin (for PostgreSQL)

Objectives

- To design and implement a PL/SQL package that includes procedures and shared cursors for structured and modular program development.

Problem Statement

- In enterprise applications, related database operations should be grouped together for better management and reuse.

Practical/Experiment Steps

- Modular Architecture Design: Structured the database logic into a two-part PL/SQL package consisting of a public specification and a private implementation body.

- Encapsulation of Global Assets: Defined a shared cursor within the package body to provide a centralized and reusable data source for multiple procedures.
- Procedural Logic Development: Authored specialized procedures to handle varied data retrieval tasks, including full table iteration and targeted record searching.
- Information Hiding Implementation: Enforced security and abstraction by hiding the internal cursor logic from the end-user while exposing only necessary procedure calls.
- Integration Testing: Executed the package subprograms within anonymous blocks to validate cross-procedure consistency and data accuracy.

Procedure

- Established a connection to the Oracle/PostgreSQL environment and enabled the output console for result verification.
- Created the employees table and populated it with a baseline dataset to facilitate testing.
- Drafted the Package Specification (emp_package AS) to declare the public interface of the procedures.
- Constructed the Package Body (emp_package AS BODY) to house the shared emp_cursor and the actual logic for the declared procedures.
- Implemented a FOR loop within show_employees to automatically handle the opening, fetching, and closing of the shared cursor.
- Developed the logic for get_employee to filter the shared cursor data based on a user-provided input parameter.
- Compiled both the specification and body to ensure there were no syntax errors or dependency issues.
- Invoked emp_package.show_employees to test global data retrieval and printed the formatted output.
- Called emp_package.get_employee(2) to verify the ability of the package to handle specific conditional requests using shared resources.

Input/Output Analysis

SQL Input Queries

```
CREATE TABLE employees (
  emp_id NUMBER PRIMARY KEY,
  emp_name VARCHAR2(50),
```

```
    salary NUMBER
);
```

Output

Query result	Script output	DBMS output	Explain Plan	SQL history
  <pre>SQL> CREATE TABLE employees (emp_id NUMBER PRIMARY KEY, emp_name VARCHAR2(50), salary NUMBER... Show more...</pre> Table EMPLOYEES created. Elapsed: 00:00:00.020				

SQL Input Queries

```
INSERT INTO employees VALUES (1, 'Amit', 30000);
INSERT INTO employees VALUES (2, 'Riya', 40000);
INSERT INTO employees VALUES (3, 'John', 50000);
```

```
COMMIT;
```

Output

Query result	Script output	DBMS output	Explain Plan	SQL history
  <pre>SQL> INSERT INTO employees VALUES (1, 'Amit', 30000) 1 row inserted. Elapsed: 00:00:00.019 SQL> INSERT INTO employees VALUES (2, 'Riya', 40000) 1 row inserted. Elapsed: 00:00:00.002 SQL> INSERT INTO employees VALUES (3, 'John', 50000) 1 row inserted. Elapsed: 00:00:00.001</pre>				

```
SQL> COMMIT
```

```
Commit complete.
```

```
Elapsed: 00:00:00.002
```

SQL Input Queries

```
CREATE OR REPLACE PACKAGE emp_package AS
```

```
-- Procedure to display all employee details
```

```
PROCEDURE show_employees;
```

```
-- Procedure to display employee by ID
```

```
PROCEDURE get_employee(p_id NUMBER);
```

```
END emp_package;
```

```
/
```

Output

Query result **Script output** DBMS output Explain Plan SQL history



```
-- Procedure to display all employee details
PROCEDURE show_employees;
...
Show more...
```

```
Package EMP_PACKAGE compiled
```

```
Elapsed: 00:00:00.011
```

SQL Input Queries

```
CREATE OR REPLACE PACKAGE BODY emp_package AS
```

```
CURSOR emp_cursor IS
```

```
SELECT emp_id, emp_name, salary FROM employees;
```

```
PROCEDURE show_employees IS
```

```
BEGIN
```

```
FOR rec IN emp_cursor LOOP
```

```
DBMS_OUTPUT.PUT_LINE('ID: ' || rec.emp_id ||
```

```
' Name: ' || rec.emp_name ||
```

```
' Salary: ' || rec.salary);
```

```
END LOOP;
```

```
END;
```

```
PROCEDURE get_employee(p_id NUMBER) IS
BEGIN
    FOR rec IN emp_cursor LOOP
        IF rec.emp_id = p_id THEN
            DBMS_OUTPUT.PUT_LINE('Employee Found -> ID: ' || rec.emp_id ||
                                  ' Name: ' || rec.emp_name ||
                                  ' Salary: ' || rec.salary);
        END IF;
    END LOOP;
END;
```

```
END emp_package;
/
```

Output

Query result **Script output** DBMS output Explain Plan SQL history



```
-- Shared cursor (used by multiple procedures)
CURSOR emp_cursor IS...
```

[Show more...](#)

Package Body EMP_PACKAGE compiled

Elapsed: 00:00:00.031

SQL Input Queries

```
SET SERVEROUTPUT ON;
```

```
BEGIN
    emp_package.show_employees;
END;
/
CALL update_student(1, 'Rahul Verma', 22, 'Data Science');
```

Output

Query result	Script output	DBMS output	Explain Plan	SQL history
 ID: 1 Name: Amit Salary: 30000 ID: 2 Name: Riya Salary: 40000 ID: 3 Name: John Salary: 50000 PL/SQL procedure successfully completed. Elapsed: 00:00:00.128				

SQL Input Queries

```
BEGIN
    emp_package.get_employee(2);
END;
/
```

Output

Query result	Script output	DBMS output	Explain Plan	SQL history
 <pre>emp_package.get_employee(2); END;</pre> Employee Found -> ID: 2 Name: Riya Salary: 40000 PL/SQL procedure successfully completed. Elapsed: 00:00:00.123				

Learning Outcomes

- **Package Structure Proficiency:** Gained a clear understanding of the separation between the package specification (header) and the package body (implementation).
- **Modular Programming Mastery:** Learned to group related database operations into a single logical unit to improve maintainability and code organization.
- **Shared Resource Management:** Expertise in utilizing shared cursors to reduce memory overhead and ensure consistent data fetching across multiple subprograms.

- Encapsulation and Security: Understood how packages provide a layer of abstraction by hiding complex implementation details from the calling environment.